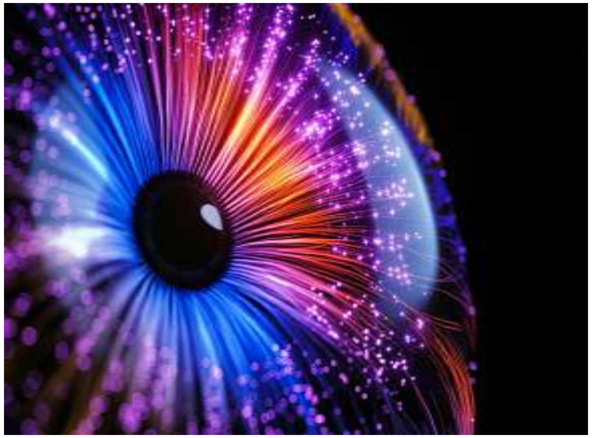
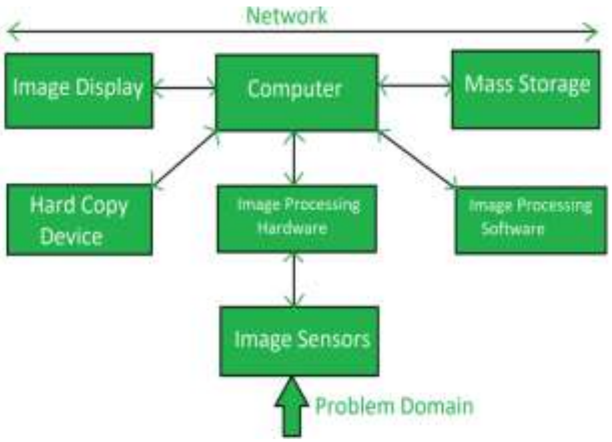
Digital Image Processing





Dr. Md. Aynal Haque

Professor, EEE, BUET

Email: aynal@eee.buet.ac.bd

Cell: 018 1763 8600

Topic 5

Image Restoration and Reconstruction

Image Restoration

- **Image restoration**: recover an image that has been degraded by using a prior knowledge of the degradation phenomenon.
- Model the degradation and applying the inverse process in order to recover the original image.

Image Restoration

Image restoration is the operation of taking a corrupt/noisy image and estimating the clean, original image. Corruption may come in many forms such as motion blur, noise and camera mis-focus. Image restoration is performed by reversing the process that blurred the image and such is performed by imaging a point source and use the point source image, to restore the image information lost to the blurring process.

Image restoration is different from image enhancement in that the latter is designed to emphasize features of the image that make the image more pleasing to the observer, but not necessarily to produce realistic data from a scientific point of view. Image enhancement techniques (like contrast stretching or de-blurring by a nearest neighbor procedure) provided by imaging packages use no *a priori* model of the process that created the image.

With image enhancement noise can effectively be removed by sacrificing some resolution, but this is not acceptable in many applications. In a fluorescence microscope, resolution in the z-direction is bad as it is. More advanced image processing techniques must be applied to recover the object.

Image Restoration

Objective: to reduce noise and recover resolution loss. Techniques are performed either in the **image domain** or the **frequency domain**. The most straightforward and a conventional technique for image restoration is deconvolution, which is performed in the frequency domain and after computing the Fourier transform of both the image and the PSF (point spread function) and undo the resolution loss caused by the blurring factors. Nowadays, photo restoration is done using digital tools and software to fix any type of damage images may have and improve the general quality and definition of the details.

Types of Corrections

1. Geometric correction
2. Radiometric correction
3. Denoising

Image restoration techniques aim to reverse the effects of degradation and restore the image as closely as possible to its original or desired state. The process involves analysing the image and applying algorithms and filters to remove or reduce the degradations. The ultimate goal is to enhance the visual quality, improve the interpretability, and extract relevant information from the image.

Image Restoration

Spatial domain methods: Spatial domain techniques primarily operate on the pixel values of an image. Some common methods in this domain include:

- **Median filtering:** This technique replaces each pixel value with the median value in its local neighborhood, effectively reducing impulse noise.
- **Wiener filtering:** Based on statistical models, the Wiener filter minimizes the mean square error between the original image and the filtered image. It is particularly useful for reducing noise and enhancing blurred images.
- **Total variation regularization:** This technique minimizes the total variation of an image while preserving important image details. It is effective in removing noise while maintaining image edges.

Image Restoration

Frequency domain methods: Frequency domain techniques involve transforming the image from the spatial domain to the frequency domain, typically using the Fourier transform. Some common methods in this domain include:

- **Inverse filtering:** This technique aims to recover the original image by estimating the inverse of the degradation function. However, it is highly sensitive to noise and can amplify noise in the restoration process.
- **Constrained least squares filtering:** By incorporating constraints on the solution, this method reduces noise and restores the image while preserving important image details.
- **Homomorphic filtering:** It is used for enhancing images that suffer from both additive and multiplicative noise. This technique separately processes the low-frequency and high-frequency components of the image to improve visibility.

Image Restoration

Application Areas: Image restoration has a wide range of applications in various fields, including:

- **Forensic analysis:** In criminal investigations, image restoration techniques can help enhance surveillance footage, recover details from low-quality images, and improve the identification of objects or individuals.
- **Medical imaging:** Image restoration is crucial in medical imaging to improve the accuracy of diagnosis. It helps in reducing noise, enhancing contrast, and improving image resolution for techniques such as X-ray, MRI, CT scans, and ultrasound.
- **Photography:** Image restoration techniques are commonly used in digital photography to correct imperfections caused by factors like motion blur, lens aberrations, and sensor noise. They can also be used to restore old and damaged photographs.
- **Archival preservation:** Image restoration plays a significant role in preserving historical documents, artworks, and photographs. By reducing noise, enhancing faded details, and removing artifacts, valuable visual content can be preserved for future generations.

Image Restoration

Challenges and future directions

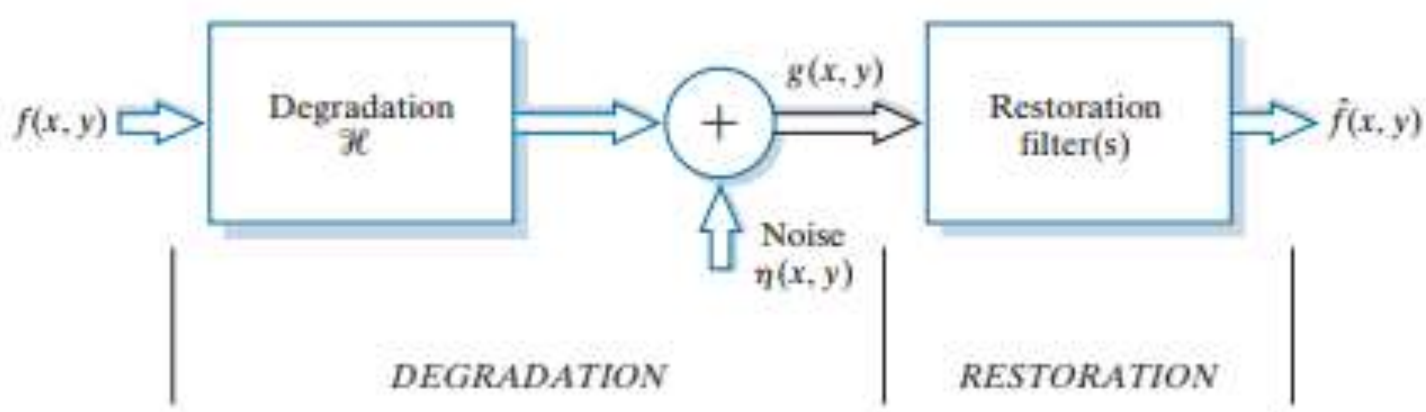
- Despite significant advancements in image restoration, several challenges remain. Some of the key challenges include handling complex degradations, dealing with limited information, and addressing the trade-off between restoration quality and computation time.

- The future of image restoration is likely to be driven by developments in deep learning and artificial intelligence. Convolutional neural networks (CNNs) have shown promising results in various image restoration tasks, including denoising, super-resolution, and inpainting. The use of generative adversarial networks (GANs) has also gained attention for realistic image restoration.

- Additionally, emerging technologies such as computational photography and multi-sensor imaging are expected to provide new avenues for image restoration research and applications.

Model of Image Degradation/Restoration

FIGURE 5.1 A model of the image degradation/ restoration process.



- Degradation
  - Degradation function H
  - Additive noise $\eta(x, y)$

Image Degradation

Image degradation is the deterioration of an image's quality due to factors such as noise, blur, and artifacts that occur during the image acquisition, processing, storage, or transmission phases. This reduction in quality can lead to visible imperfections, loss of detail, color inaccuracies, and distortions, impacting the image's overall appearance and its usefulness in subsequent tasks like image recognition.

Common Causes of Image Degradation

- **Noise**: Random fluctuations in pixel values, which can be introduced by the image sensor itself or during transmission.
- **Blur**: Can result from misfocus, object motion, camera shake, or atmospheric turbulence.
- **Artifacts**: Unwanted variations in pixel intensity or color, often introduced during image compression or transmission.
- **Distortion**: Geometric changes to the image, such as pincushion or barrel distortion, can affect the image's structure.
- **Color Inaccuracies**: Problems with color balance or saturation, leading to incorrect color representation.
- **Lossy Compression**: The process of reducing file size by discarding some image data, which can lead to a loss of detail and quality.

Image Degradation

Impact of Image Degradation

- **Reduced Visual Quality**: The most direct impact is a noticeable decline in the image's clarity, detail, and aesthetic appeal.
- **Compromised Subsequent Operations**: Degraded images can negatively affect the performance of image analysis tasks, such as image recognition, object detection, or medical diagnosis, as these systems rely on clear input data.
- **Loss of Information**: Degradation can cause a loss of original scene information, making it harder to interpret the content of the image.

Image Degradation

If H is a linear, position-invariant process, then the degraded image is given in the spatial domain by

$$g(x,y) = h(x,y) \star f(x,y) + \eta(x,y)$$

The model of the degraded image is given in the frequency domain by

$$G(u,v) = H(u,v)F(u,v) + N(u,v)$$

Noise Sources

- The principal sources of noise in digital images arise during **image acquisition and/or transmission**

✓ Image acquisition

e.g., light levels, sensor temperature, etc.

✓ Transmission

e.g., lightning or other atmospheric disturbance in wireless network

Noise Models

- White noise
  - The Fourier spectrum of noise is constant
- With the exception of spatially periodic noise, we assume
  - Noise is independent of spatial coordinates
  - Noise is uncorrelated with respect to the image itself

➢ **Gaussian noise**

Electronic circuit noise, sensor noise due to poor illumination and/or high temperature

➢ **Rayleigh noise**

Range imaging

Range Imaging

- Short for High Dynamic Range Imaging. HDRI is an imaging technique that allows for a greater dynamic range of exposure than would be obtained through any normal imaging process.

- It is now popularly used to refer to the process of tone mapping together with bracketed exposures of normal digital images, giving the end result a high, often exaggerated dynamic range

Tone mapping is a technique used in image processing and computer graphics to map a set of colors to another; often to approximate the appearance of HDRI in media with a more limited dynamic range

Bracketing is the general technique of taking several shots of the same subject using different or the same camera settings

- The intention of HDRI is to accurately represent the wide range of intensity levels found in real scenes ranging from direct sunlight to shadows
- HDRI, also called HDR (High Dynamic Range) is a feature commonly found in high-end graphics and imaging software

Range Imaging: Example

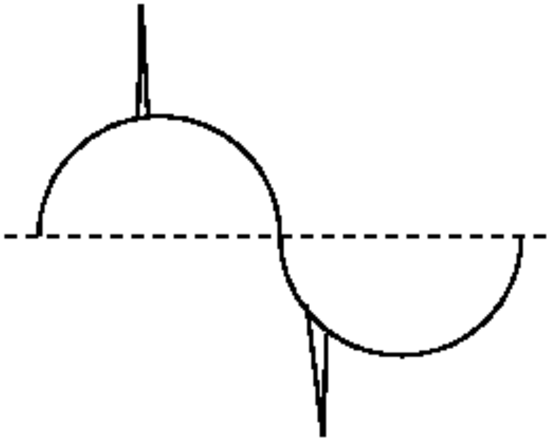
Sydney Harbor Bridge HDRI produces greater detail and fewer shadows

Range Imaging: Example

Tower Bridge in Sacramento, CA

Noise Models

- **Erlang (gamma) noise**: Laser imaging
- **Exponential noise**: Laser imaging
- **Uniform noise**: Least descriptive; Basis for numerous random number generators
- **Impulse noise**: quick transients, such as faulty switching



Impulse Noise

Gaussian Noise

The PDF of Gaussian random variable, z, is given by

$$p(z) = \frac{1}{\sqrt{2\pi\sigma}} e^{-(z-\bar{z})^2/2\sigma^2}$$

where, z represents intensity

$\bar{z}$ is the mean (average) value of z

σ is the standard deviation

- 70% of its values will be in the range $\left[(\mu-\sigma),(\mu+\sigma)\right]$
- 95% of its values will be in the range $\left[(\mu-2\sigma),(\mu+2\sigma)\right]$

Rayleigh Noise

The PDF of Rayleigh noise is given by

$$p(z) = \begin{cases} \dfrac{2}{b}(z-a)e^{-(z-a)^2/b} & \text{for } z \geq a \\ 0 & \text{for } z < a \end{cases}$$

The mean and variance of this density are given by

$$\bar{z} = a + \sqrt{\pi b / 4}$$

$$\sigma^2 = \frac{b(4-\pi)}{4}$$

Erlang (Gamma) Noise

The PDF of Erlang noise is given by

$$p(z) = \begin{cases} \dfrac{a^b z^{b-1}}{(b-1)!} e^{-az} & \text{for } z \geq 0 \\ 0 & \text{for } z < a \end{cases}$$

The mean and variance of this density are given by

$$\bar{z} = b / a$$

$$\sigma^2 = b / a^2$$

Exponential Noise

The PDF of exponential noise is given by

$$p(z) = \begin{cases} ae^{-az} & \text{for } z \geq 0 \\ 0 & \text{for } z < a \end{cases}$$

The mean and variance of this density are given by

$$\bar{z} = 1/a$$

$$\sigma^2 = 1/a^2$$

Uniform Noise

The PDF of uniform noise is given by

$$p(z) = \begin{cases} \dfrac{1}{b-a} & \text{for a} \leq z \leq b \\ 0 & \text{otherwise} \end{cases}$$

The mean and variance of this density are given by

$$\bar{z} = (a+b)/2$$

$$\sigma^2 = (b-a)^2/12$$

Impulse (Salt-and-Pepper) Noise

The PDF of (bipolar) impulse noise is given by

$$p(z) = \begin{cases} P_a & \text{for } z = a \\ P_b & \text{for } z = b \\ 0 & \text{otherwise} \end{cases}$$

if $b > a$, gray-level b will appear as a light dot, while level a will appear like a dark dot.

If either P_a or P_b is zero, the impulse noise is called *unipolar*

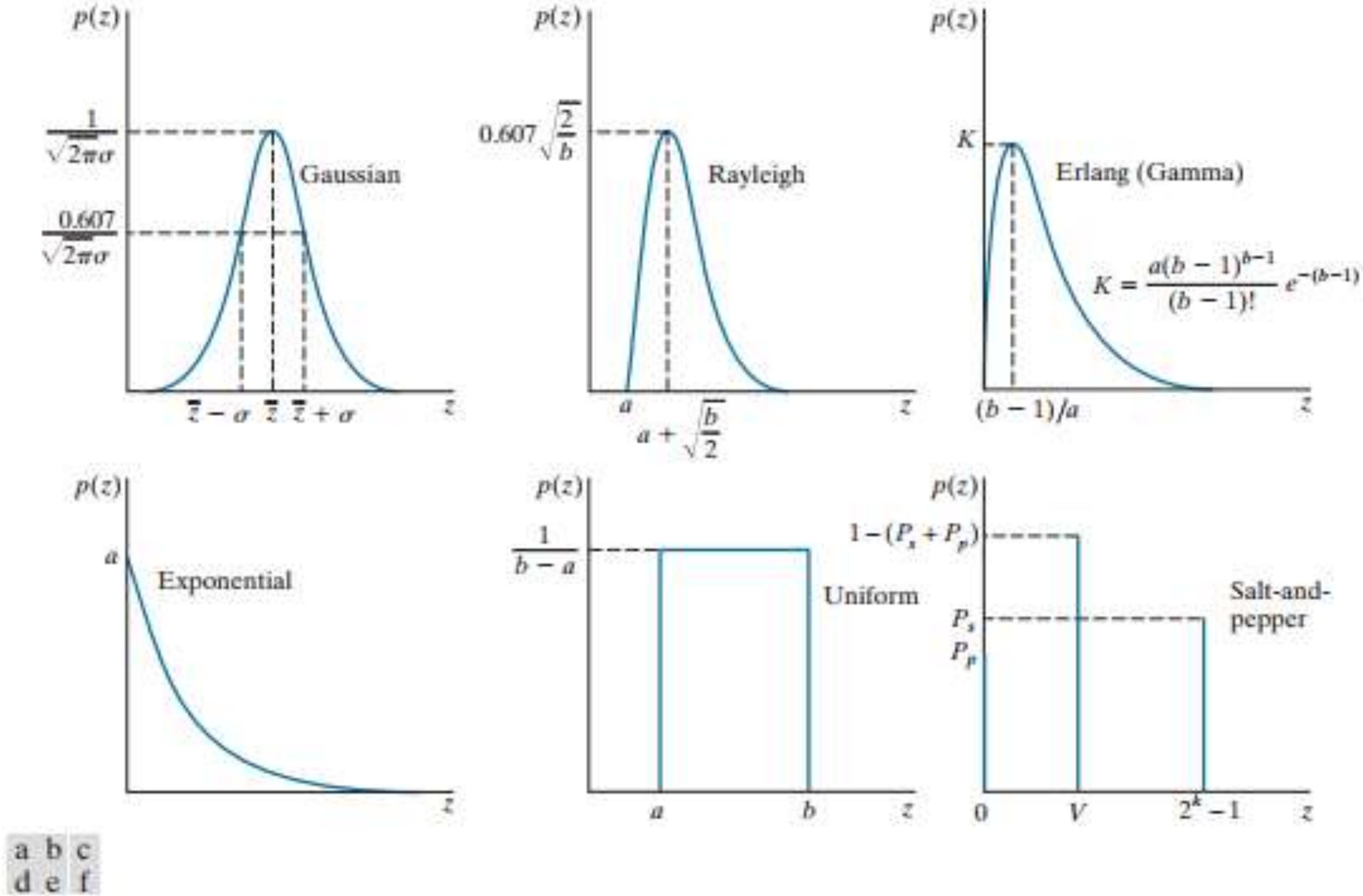


FIGURE 5.2 Some important probability density functions.

Examples of Noise

Original Image

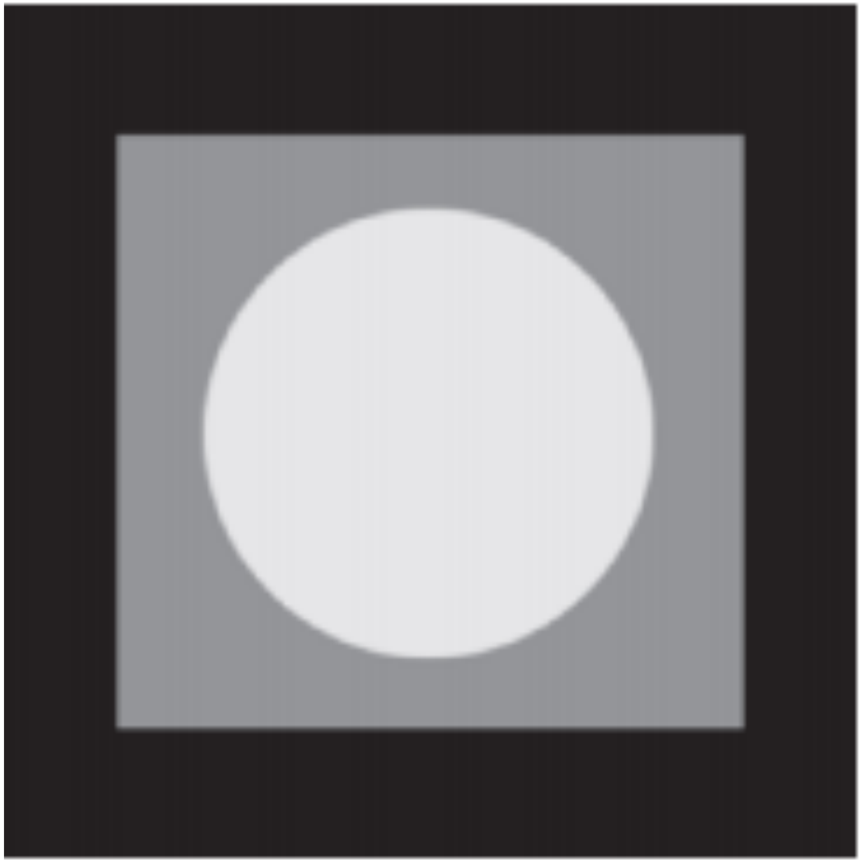
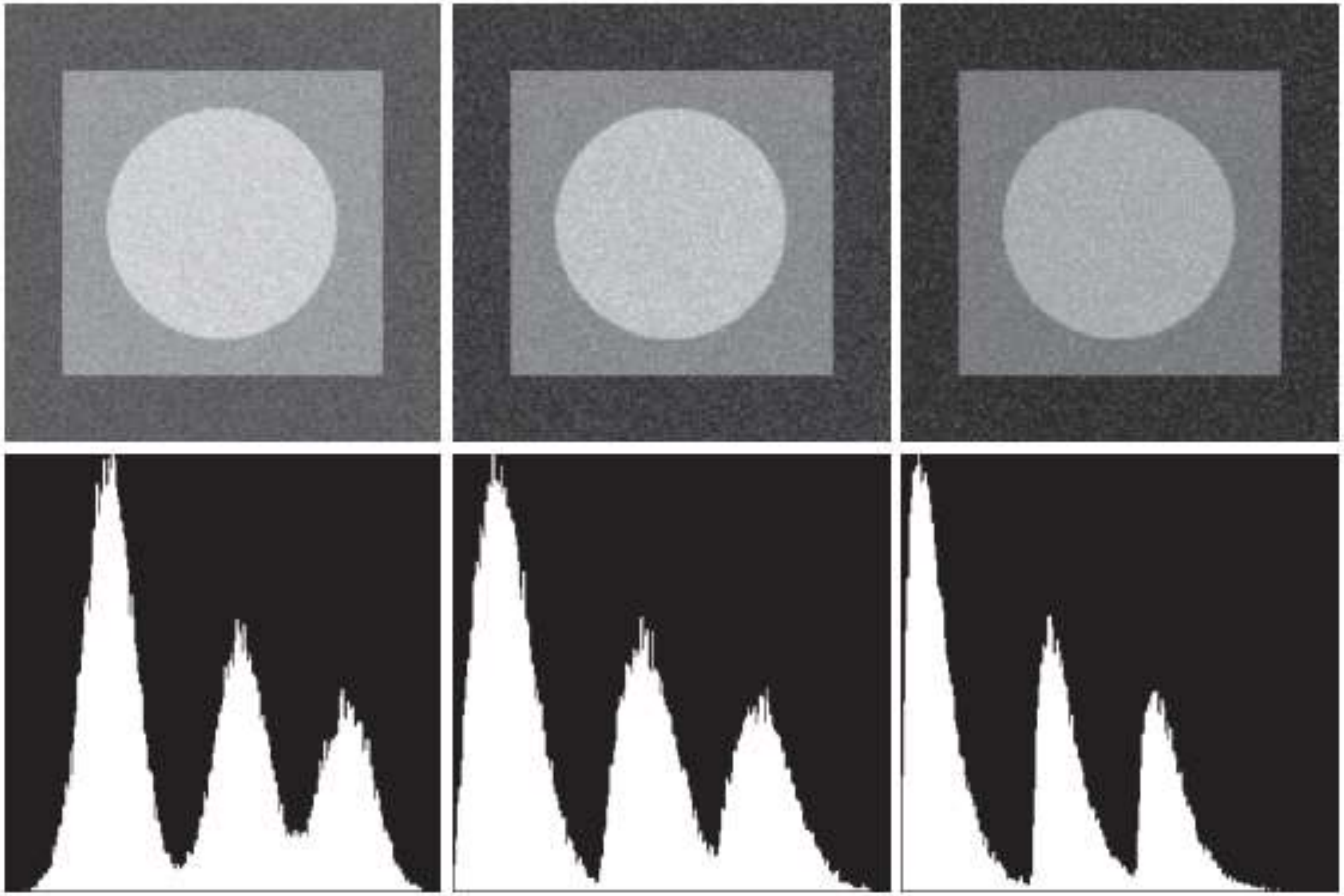


FIGURE 5.3
Test pattern used to illustrate the characteristics of the PDFs from Fig. 5.2.

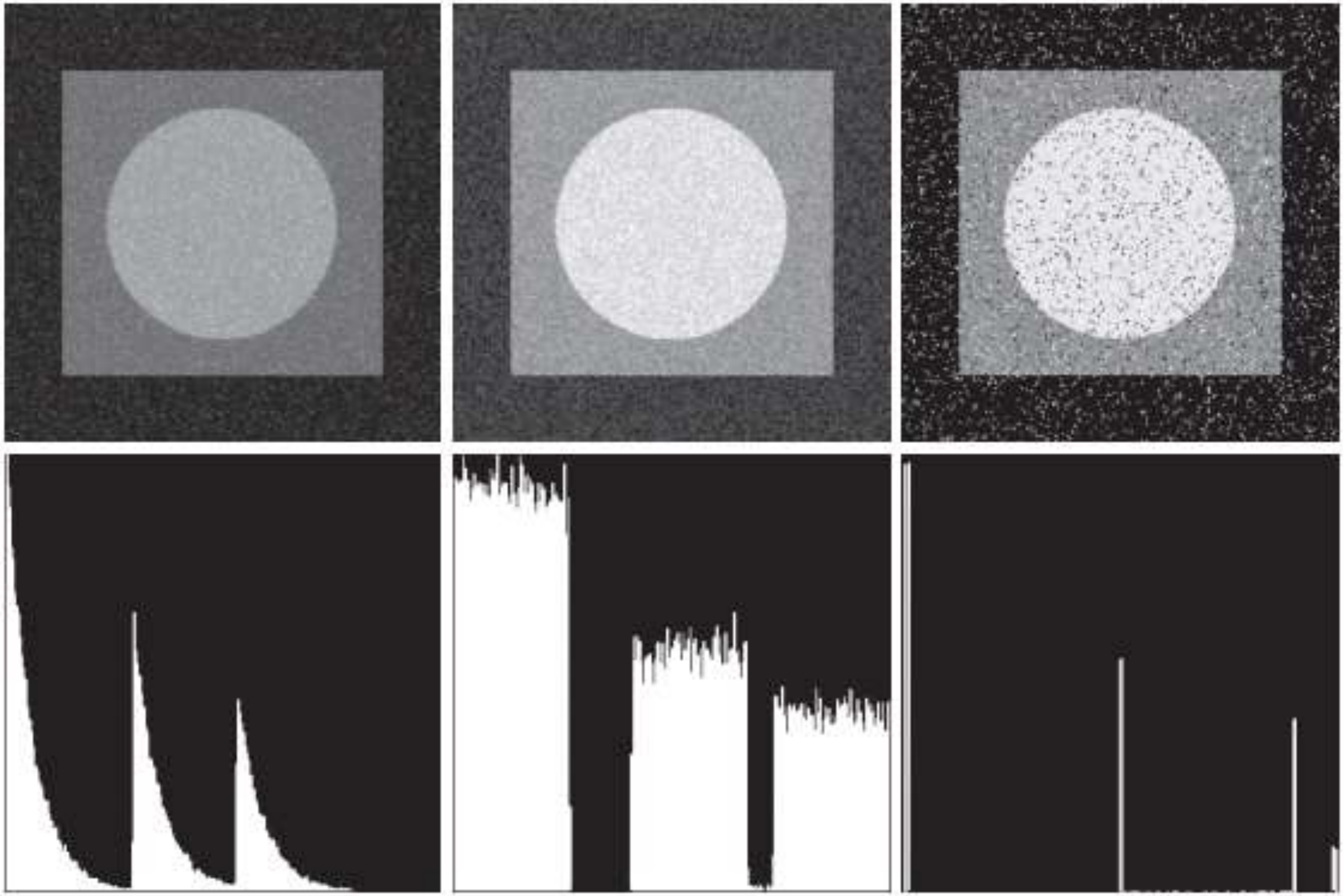
Noisy Images



a b c
d e f

FIGURE 5.4 Images and histograms resulting from adding Gaussian, Rayleigh, and Erlanga noise to the image in Fig. 5.3.

Noisy Images



g	h	i
j	k	l

FIGURE 5.4 (*continued*) Images and histograms resulting from adding exponential, uniform, and salt-and-pepper noise to the image in Fig. 5.3. In the salt-and-pepper histogram, the peaks in the origin (zero intensity) and at the far end of the scale are shown displaced slightly so that they do not blend with the page background.

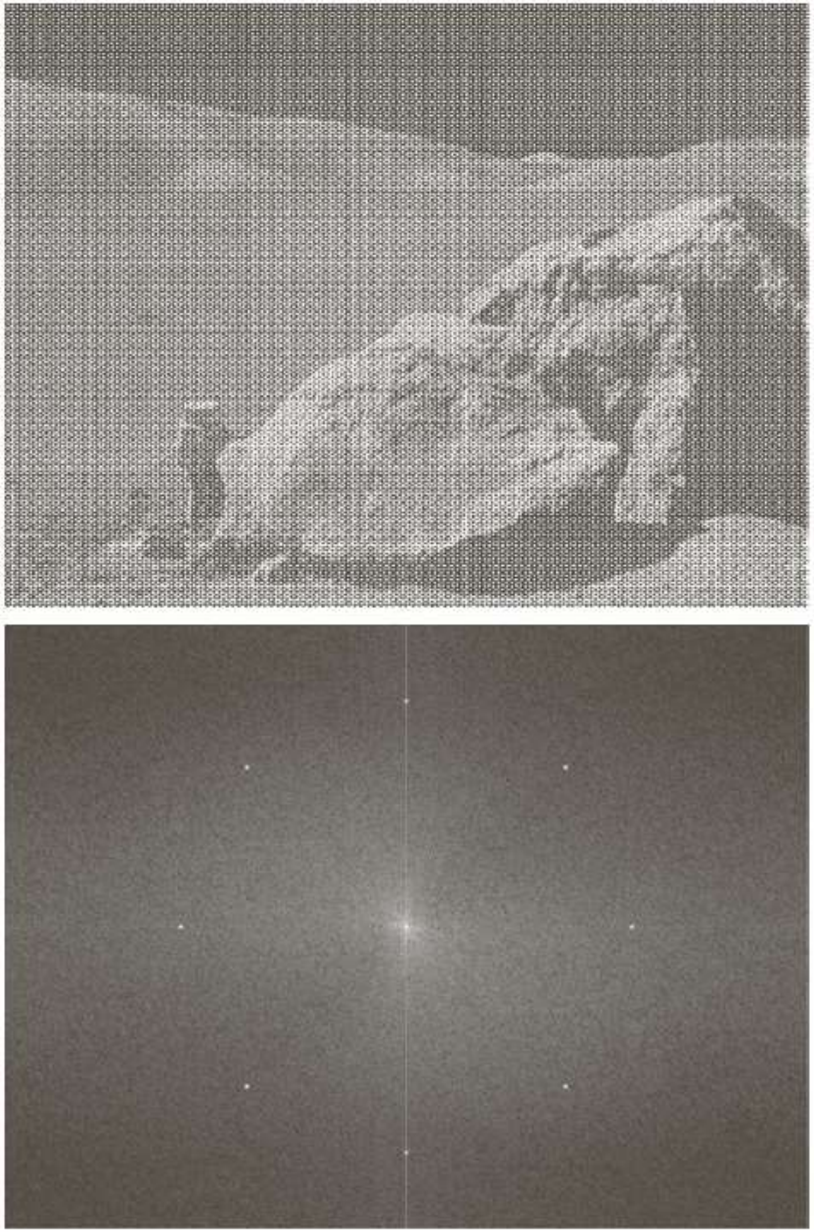
Periodic Noise

- Periodic noise in an image arises typically from electrical or electromechanical interference during image acquisition.

- It is a type of spatially dependent noise

- Periodic noise can be reduced significantly via frequency domain filtering

Example

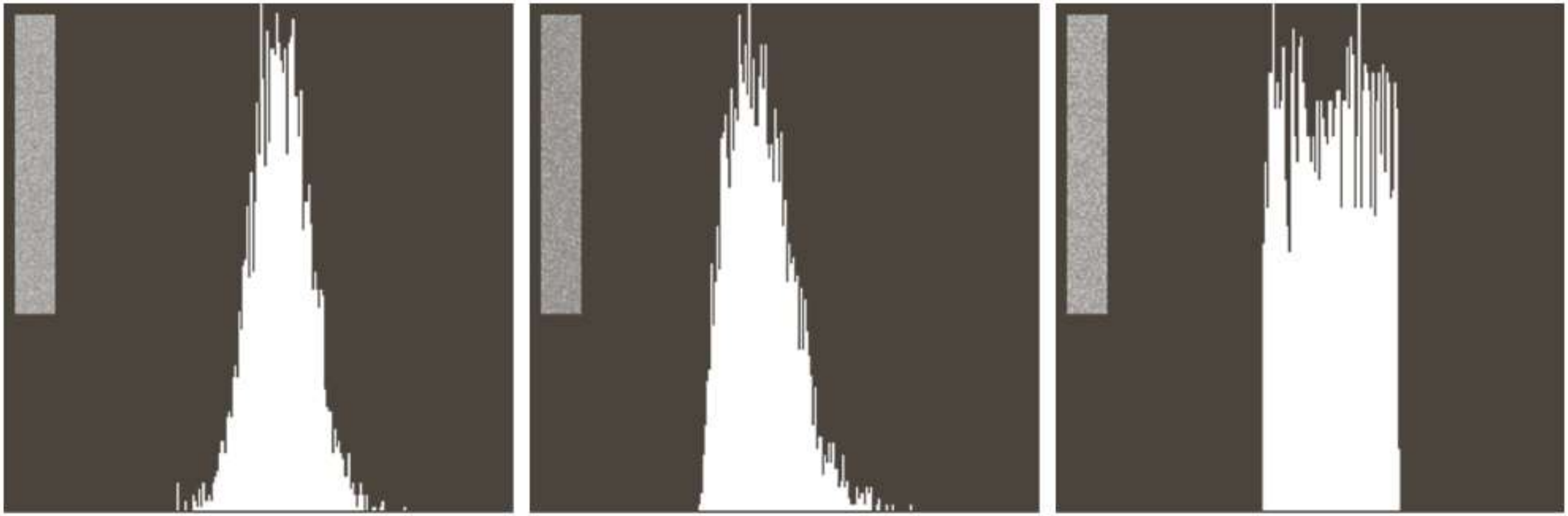


a
b

FIGURE 5.5
(a) Image corrupted by sinusoidal noise. (b) Spectrum (each pair of conjugate impulses corresponds to one sine wave). (Original image courtesy of NASA.)

Estimation of Noise Parameters

The shape of the histogram identifies the closest PDF match



a b c

FIGURE 5.6 Histograms computed using small strips (shown as inserts) from (a) the Gaussian, (b) the Rayleigh, and (c) the uniform noisy images in Fig. 5.4.

Estimation of Noise Parameters

Consider a subimage denoted by S, and let $p_s(z_i)$, $i = 0,\ 1,\ \ldots,\ L-1$, denote the probability estimates of the intensities of the pixels in S. The mean and variance of the pixels in S:

$$\bar{z} = \sum_{i=0}^{L-1} z_i p_s(z_i)$$

and

$$\sigma^2 = \sum_{i=0}^{L-1} (z_i - \bar{z})^2 p_s(z_i)$$

Image Restoration

Image Restoration in the presence of noise only: **Spatial Filtering**

Noise model without degradation

$$g(x, y) = f(x, y) + \eta(x, y)$$

and

$$G(u, v) = F(u, v) + N(u, v)$$

Spatial Filtering: Mean Filters

Let S_{xy} represent the set of coordinates in a rectangle subimage window of size $m \times n$, centered at (x, y).

Arithmetic mean filter

$$f(x,y) = \frac{1}{mn} \sum_{(s,t)\in S_{xy}} g(s,t)$$

Geometric mean filter

$$f(x,y) = \left[\prod_{(s,t)\in S_{xy}} g(s,t) \right]^{\frac{1}{mn}}$$

Generally, a geometric mean filter achieves smoothing comparable to the arithmetic mean filter, but it tends to lose less image detail in the process

Spatial Filtering: Mean Filters

Harmonic mean filter

$$f(x,y) = \left. \frac{mn}{\displaystyle\sum_{(s,t)\in S_{xy}} \frac{1}{g(s,t)}} \right]^{\frac{1}{mn}}$$

It works well for salt noise, but fails for pepper noise.
It does well also with other types of noise like Gaussian noise.

Spatial Filtering: Mean Filters

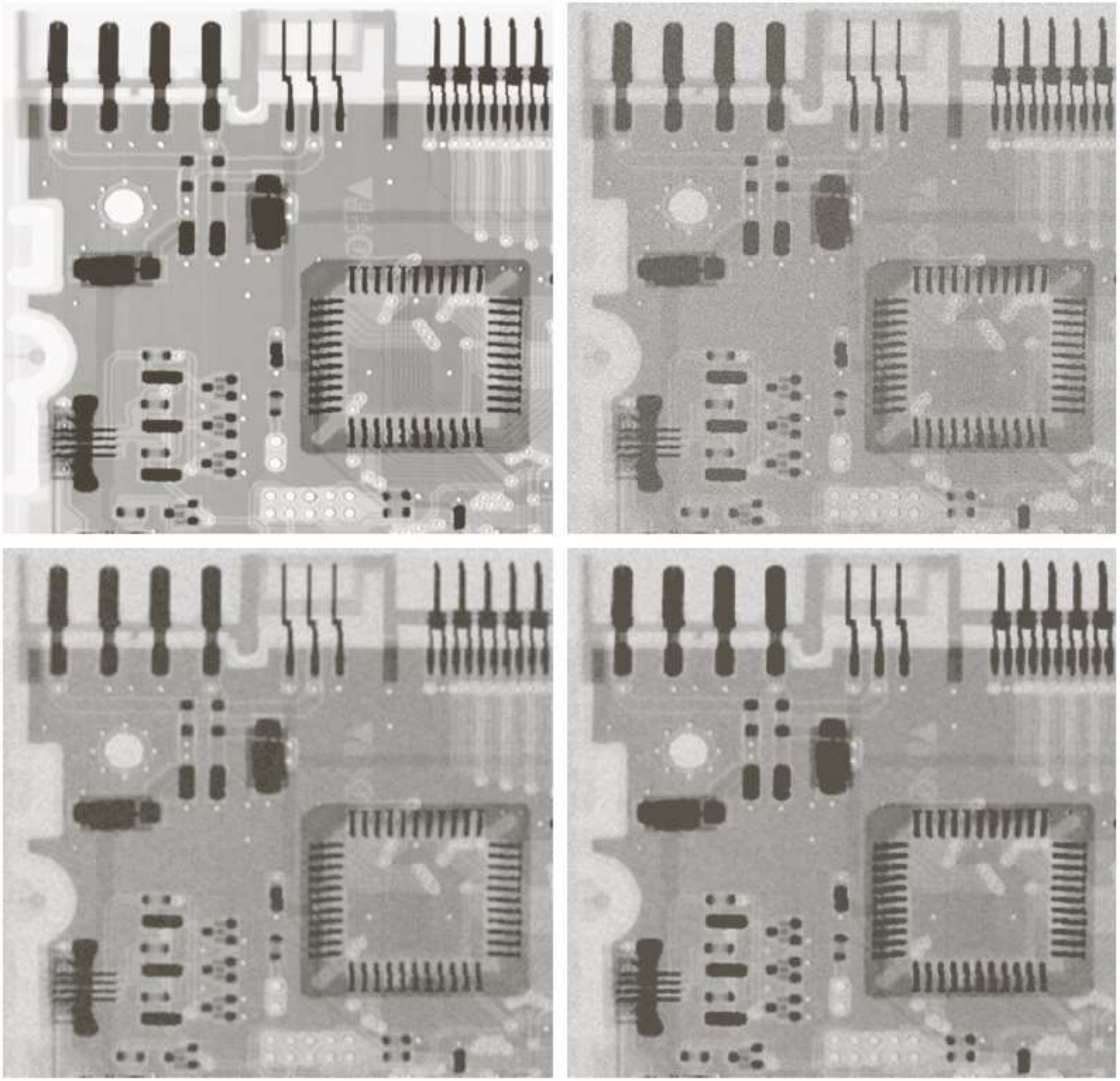
Contraharmonic mean filter

$$\hat{f}(x,y)=\frac{\sum_{(s,t)\in S_{xy}} g(s,t)^{Q+1}}{\sum_{(s,t)\in S_{xy}} g(s,t)^{Q}}$$

Q is the order of the filter.

It is well suited for reducing the effects of salt-and-pepper noise.
Q>0 for pepper noise and Q<0 for salt noise.

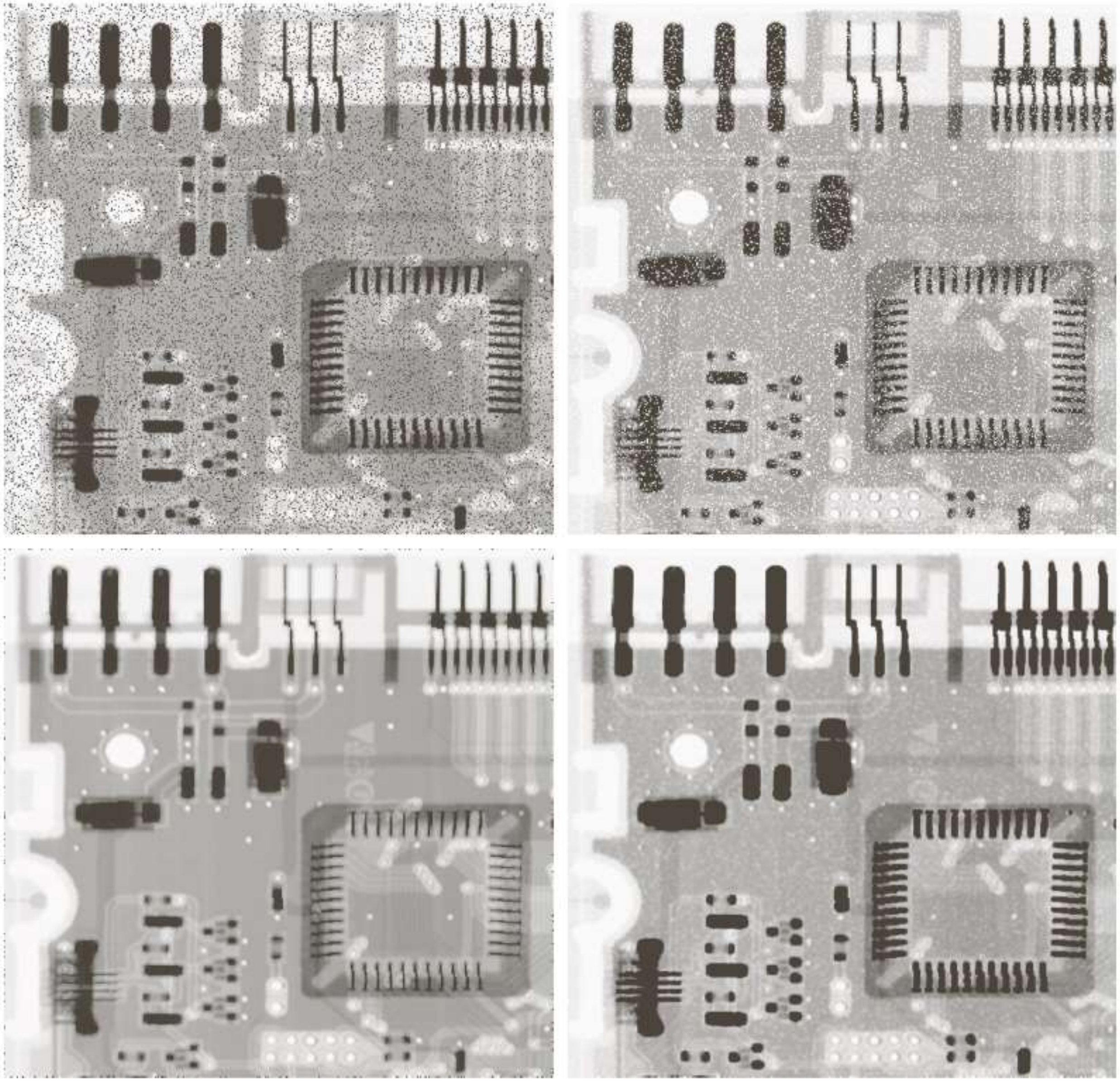
Spatial Filtering: Example



a b
c d

FIGURE 5.7 (a) X-ray image. (b) Image corrupted by additive Gaussian noise. (c) Result of filtering with an arithmetic mean filter of size 3×3. (d) Result of filtering with a geometric mean filter of the same size. (Original image courtesy of Mr. Joseph E. Pascente, Lixi, Inc.)

Spatial Filtering: Example



a b
c d

FIGURE 5.8 (a) Image corrupted by pepper noise with a probability of 0.1. (b) Image corrupted by salt noise with the same probability. (c) Result of filtering (a) with a 3×3 contra-harmonic filter of order 1.5. (d) Result of filtering (b) with $Q = -1.5$.

Spatial Filtering: Example

a b

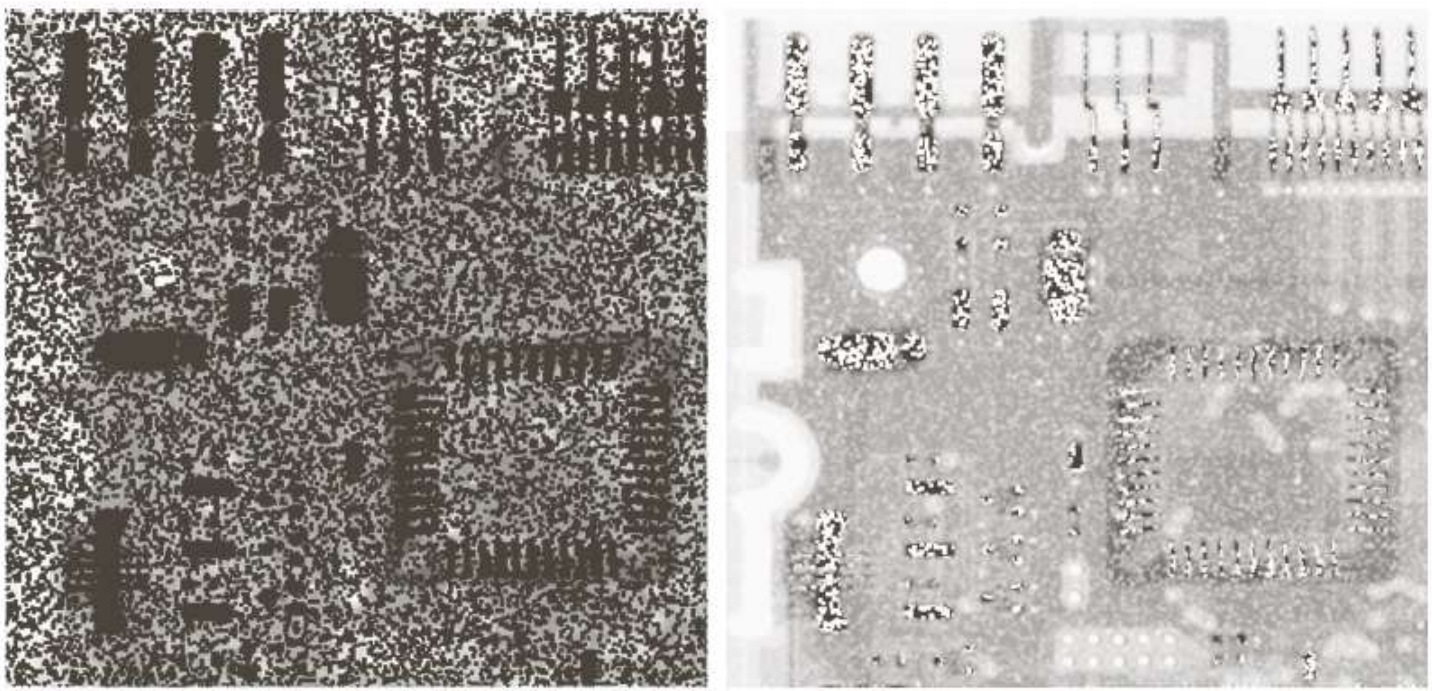


FIGURE 5.9 Results of selecting the wrong sign in contraharmonic filtering. (a) Result of filtering Fig. 5.8(a) with a contraharmonic filter of size 3×3 and $Q = -1.5$. (b) Result of filtering 5.8(b) with $Q = 1.5$.

Spatial Filtering: Order Statistic Filters

Median filter

$$f(x,y) = \underset{(s,t)\in S_{xy}}{median}\left\{g(s,t)\right\}$$

Max filter

$$f(x,y) = \max_{(s,t)\in S_{xy}}\left\{g(s,t)\right\}$$

Min filter

$$f(x,y) = \min_{(s,t)\in S_{xy}}\left\{g(s,t)\right\}$$

Midpoint filter

$$f(x,y) = \frac{1}{2}\left[\max_{(s,t)\in S_{xy}}\left\{g(s,t)\right\} + \min_{(s,t)\in S_{xy}}\left\{g(s,t)\right\}\right]$$

Spatial Filtering: Order Statistics Filters

Alpha-trimmed mean filter

$$f(x,y)=\frac{1}{mn-d}\sum_{(s,t)\in S_{xy}}\{g_r(s,t)\}$$

We delete the $d/2$ lowest and the $d/2$ highest intensity values of $g(s,t)$ in the neighborhood S_{xy}. Let $g_r(s,t)$ represent the remaining $mn - d$ pixels.

Noise Reduction

Periodic Noise Reduction by Frequency Domain Filtering

The basic idea

Periodic noise appears as concentrated bursts of energy in the Fourier transform, at locations corresponding to the frequencies of the periodic interference

Approach

A selective filter is used to isolate the noise

Bandreject Filters (Notch)

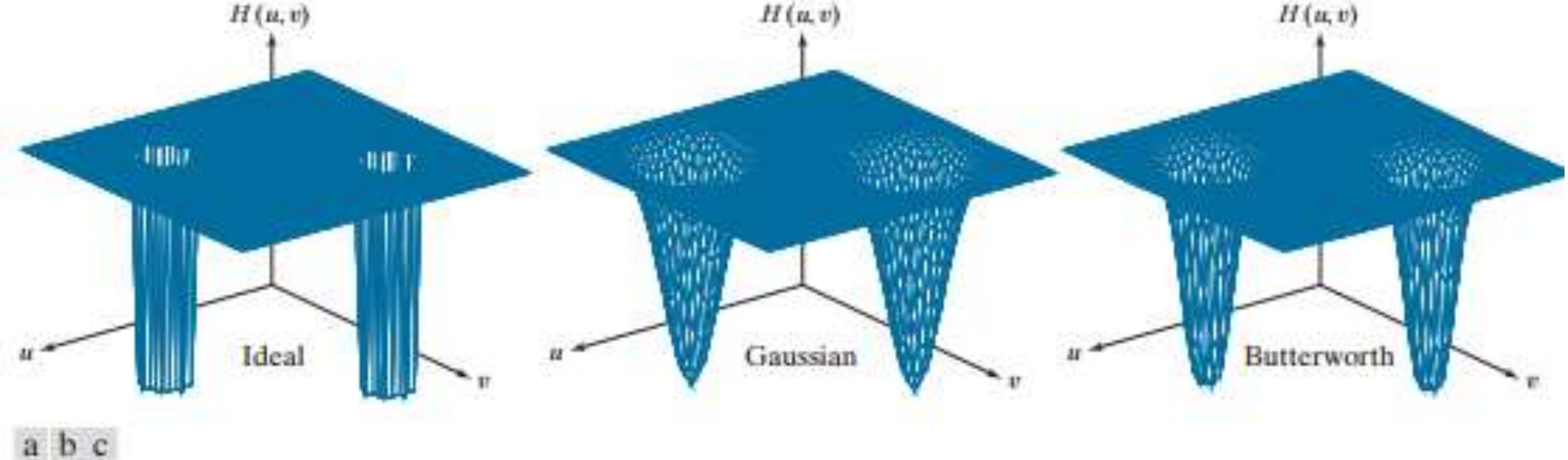


FIGURE 5.15 Perspective plots of (a) ideal, (b) Gaussian, and (c) Butterworth notch reject filter transfer functions.

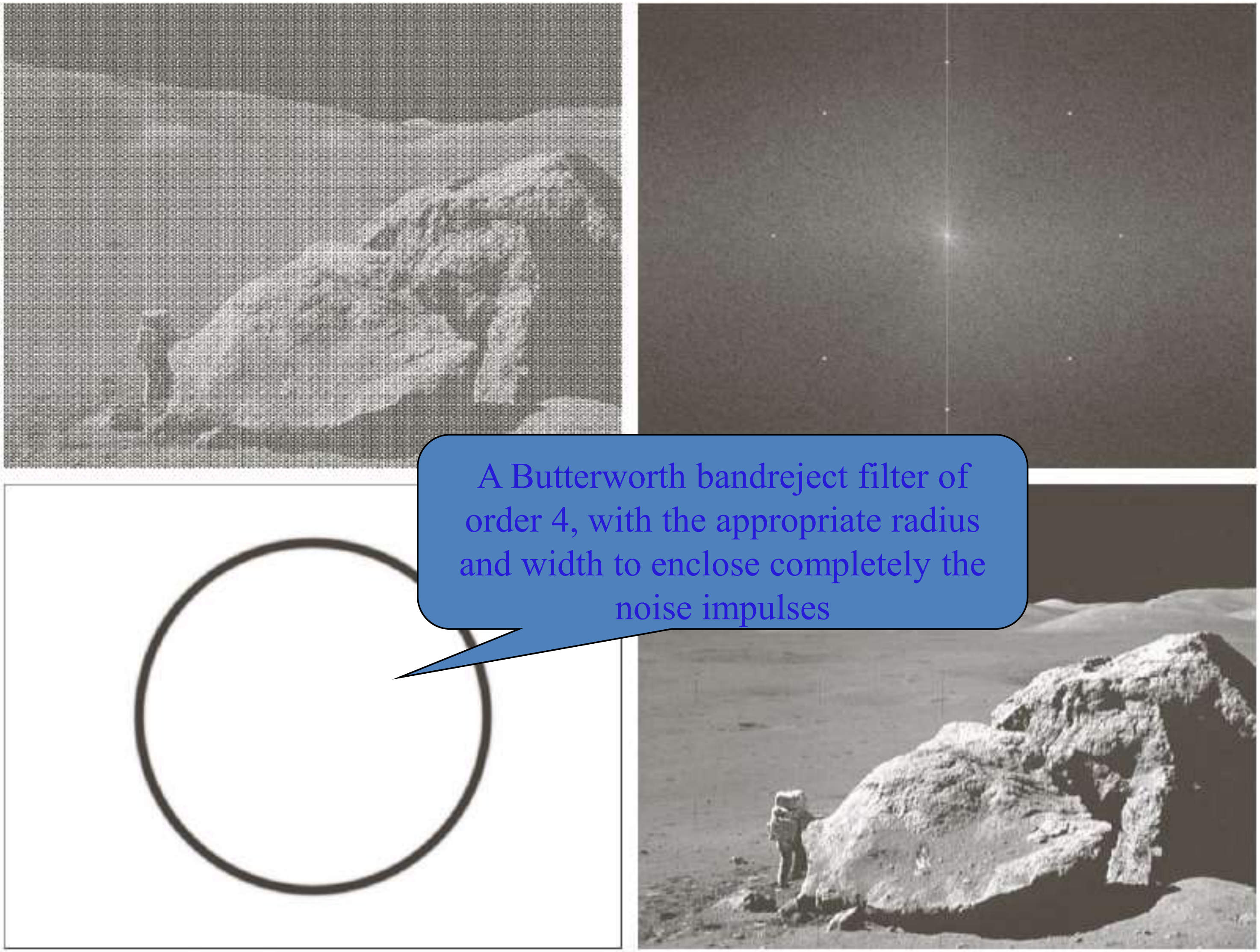


FIGURE 5.16 (a) Image corrupted by sinusoidal noise. (b) Spectrum of (a). (c) Butterworth bandreject filter (white represents 1). (d) Result of filtering. (Original image courtesy of NASA.)

Optimum Notch Filtering

It minimizes local variances of the restored estimated

$$\hat{f}(x, y)$$

Procedure for restoration tasks in multiple periodic interference

Isolate the principal contributions of the interference pattern

Subtract a variable, weighted portion of the pattern from the corrupted image

Step 1

Extract the principal frequency components of the interference pattern

Place a notch pass filter at the location of each spike.

$$N(u,v) = H_{NP}(u,v)G(u,v)$$

$$\eta(x,y) = \Im^{-1}\left\{H_{NP}(u,v)G(u,v)\right\}$$

Step 2

Filtering procedure usually yields only an approximation of the true pattern. The effect of components not present in the estimate of $\eta(x, y)$ can be minimized instead by subtracting from $g(x, y)$ a weighted portion of $\eta(x, y)$ to obtain an estimate of $f(x, y)$:

$$\hat{f}(x, y) = g(x, y) - w(x, y)\eta(x, y)$$

One approach is to select $w(x, y)$ so that the variance of the estimate $\hat{f}(x, y)$ is minimized over a specified neighborhood of every point (x, y).

The local variance of $\hat{f}(x, y)$:

$$\sigma^2(x, y) = \frac{1}{(2a+1)(2b+1)} \sum_{s=-a}^{a} \sum_{t=-b}^{b} \left[\hat{f}(x+s, y+t) - \overline{\hat{f}}(x, y) \right]^2$$

Step 3

The local variance of $f(x, y)$:

$$\sigma^2(x, y) = \frac{1}{(2a+1)(2b+1)} \sum_{s=-a}^{a} \sum_{t=-b}^{b} \left[f(x+s, y+t) - \overline{f}(x, y) \right]^2$$

$$= \frac{1}{(2a+1)(2b+1)} \sum_{s=-a}^{a} \sum_{t=-b}^{b} \left\{ \begin{array}{l} \left[g(x+s, y+t) - w(x+s, y+t)\eta(x+s, y+s) \right] \\ - \left[\overline{g}(x, y) - \overline{w(x, y)\eta(x, y)} \right] \end{array} \right\}^2$$

$$= \frac{1}{(2a+1)(2b+1)} \sum_{s=-a}^{a} \sum_{t=-b}^{b} \left\{ \begin{array}{l} \left[g(x+s, y+t) - w(x, y)\eta(x+s, y+s) \right] \\ - \left[\overline{g}(x, y) - w(x, y)\overline{\eta(x, y)} \right] \end{array} \right\}^2$$

Step 4

The local variance of $f(x, y)$:

$$\sigma^2(x,y) = \frac{1}{(2a+1)(2b+1)} \sum_{s=-a}^{a} \sum_{t=-b}^{b} \left\{ \begin{array}{l} \left[\, g(x+s, y+t) - w(x,y)\eta(x+s, y+s) \right] \\ - \left[\bar{g}(x,y) - w(x,y)\overline{\eta(x,y)} \right] \end{array} \right\}^2$$

To minimize $\sigma^2(x,y)$, $\dfrac{\partial \sigma^2(x,y)}{\partial w(x,y)} = 0$

for $w(x,y)$, the result is

$$w(x,y) = \frac{\overline{g(x,y)\eta(x,y)} - \bar{g}(x,y)\bar{\eta}(x,y)}{\overline{\eta^2}(x,y) - \bar{\eta}^2(x,y)}$$

Linear, Position-Invariant Degradations

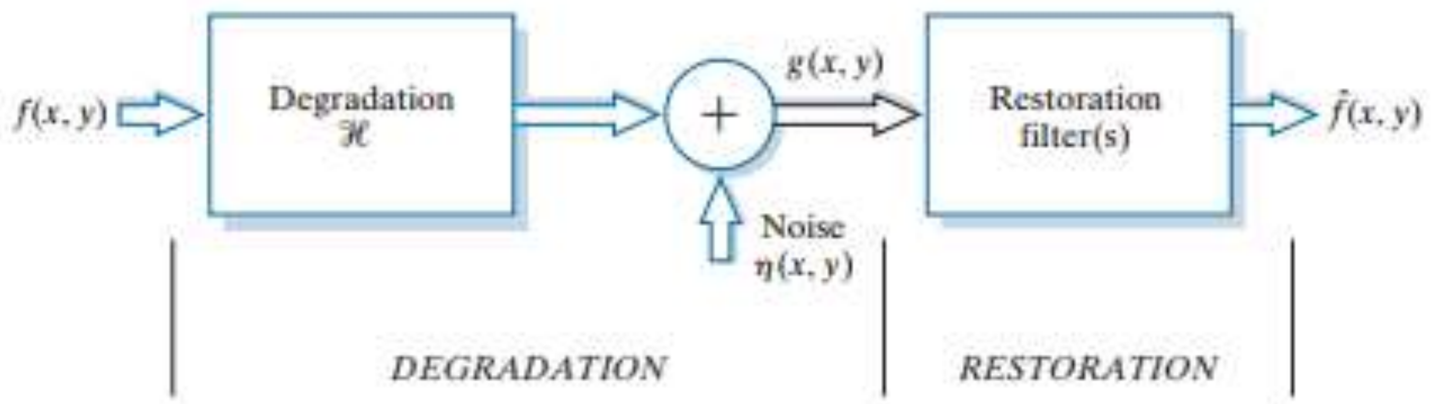


FIGURE 5.1 A model of the image degradation/ restoration process.

$$g(x,y) = H\left[f(x,y)\right] + \eta(x,y)$$

Linear, Position-Invariant Degradations

H is linear

$$H\left[af_1(x,y)+bf_2(x,y)\right]=aH\left[f_1(x,y)\right]+bH\left[f_2(x,y)\right]$$

f_1 and f_2 are any two input images.

An operator having the input-output relationship

$g(x,y)=H\left[f(x,y)\right]$ is said to be position invariant

if

$$H\left[f(x-\alpha,y-\beta)\right]=g(x-\alpha,y-\beta)$$

for any $f(x,y)$ and any α and β.

Linear, Position-Invariant Degradations

$$f(x,y)=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(\alpha,\beta)\delta(x-\alpha,y-\beta)\,d\alpha\,d\beta$$

Assume for a moment that $\eta(x,y)=0$

if H is a linear operator,

$$g(x,y)=H\left[f(x,y)\right]$$

$$=H\left[\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(\alpha,\beta)\delta(x-\alpha,y-\beta)\,d\alpha\,d\beta\right]$$

$$=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} H\left[f(\alpha,\beta)\delta(x-\alpha,y-\beta)\right]d\alpha\,d\beta$$

$$=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(\alpha,\beta)\underbrace{H\left[\delta(x-\alpha,y-\beta)\right]}_{\text{Impulse response}}d\alpha\,d\beta$$

Superposition (or Fredholm) integral of the first kind

Impulse response

Linear, Position-Invariant Degradations

Assume for a moment that $\eta(x, y) = 0$

if H is a linear operator and position invariant,

$$H\left[\delta(x-\alpha, y-\beta)\right] = h(x-\alpha, y-\beta)$$

$$g(x, y) = H\left[f(x, y)\right]$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(\alpha, \beta) H\left[\delta(x-\alpha, y-\beta)\right] d\alpha\, d\beta$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(\alpha, \beta) h(x-\alpha, y-\beta)\, d\alpha\, d\beta$$

Convolution integral in 2-D

Linear, Position-Invariant Degradations

In the presence of additive noise,

if H is a linear operator and position invariant,

$$g(x, y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(\alpha, \beta)h(x-\alpha, y-\beta)\,d\alpha\, d\beta + \eta(x, y)$$

$$= h(x, y)\bigstar f(x, y) + \eta(x, y)$$

$$G(u,v) = H(u,v)F(u,v) + N(u,v)$$

Estimating the Degradation Function

- Three principal ways to estimate the degradation function

1. Observation

2. Experimentation

3. Mathematical Modeling

Mathematical Modeling

- Environmental conditions cause degradation

A model about atmospheric turbulence

$$H(u,v) = e^{-k(u^2+v^2)^{5/6}}$$

k : a constant that depends on

the nature of the turbulence

a b
c d

FIGURE 5.25 Illustration of the atmospheric turbulence model. (a) Negligible turbulence. (b) Severe turbulence, $k = 0.0025$. (c) Mild turbulence, $k = 0.001$. (d) Low turbulence, $k = 0.00025$. (Original image courtesy of NASA.)

Mathematical Modeling

- Derive a mathematical model from basic principles

e.g., An image blurred by uniform linear motion between the image and the sensor during image acquisition

Suppose that an image $f(x, y)$ undergoes planar motion, $x_0(t)$ and $y_0(t)$ are the time-varying components of motion in the x- and y-directions, respectively.

The optical imaging process is perfect. T is the duration of the exposure. The blurred image $g(x, y)$

$$g(x, y) = \int_0^T f\left[x - x_0(t), y - y_0(t)\right] dt$$

Mathematical Modeling

$$g(x,y)=\int_0^T f\left[x-x_0(t),y-y_0(t)\right]dt$$

$$G(u,v)=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y)e^{-j2\pi(ux+vy)}dxdy$$

$$=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\left[\int_0^T f\left[x-x_0(t),y-y_0(t)\right]dt\right]e^{-j2\pi(ux+vy)}dxdy$$

$$=\int_0^T\left[\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f\left[x-x_0(t),y-y_0(t)\right]e^{-j2\pi(ux+vy)}dxdy\right]dt$$

$$=\int_0^T F(u,v)e^{-j2\pi[ux_0(t)+vy_0(t)]}dt$$

$$=F(u,v)\int_0^T e^{-j2\pi[ux_0(t)+vy_0(t)]}dt$$

Mathematical Modeling

$$H(u,v) = \int_0^T e^{-j2\pi[ux_0(t)+vy_0(t)]}dt$$

Suppose that the image undergoes uniform linear motion in the x-direction only, at a rate given by $x_0(t) = at/T$.

$$H(u,v) = \int_0^T e^{-j2\pi ux_0(t)}dt$$

$$= \int_0^T e^{-j2\pi uat/T}dt$$

$$= \frac{T}{\pi ua}\sin(\pi ua)e^{-j\pi ua}$$

Mathematical Modeling

Suppose that the image undergoes uniform linear motion in the *x*-direction and *y*-direction, at a rate given by $x_0(t) = at/T$ and $y_0(t) = bt/T$

$$H(u,v) = \int_0^T e^{-j2\pi[ux_0(t)+vy_0(t)]}dt$$

$$= \int_0^T e^{-j2\pi[ua+vb]t/T}dt$$

$$= \frac{T}{\pi(ua+vb)}\sin\left[\pi(ua+vb)\right]e^{-j\pi(ua+vb)}$$

Inverse Filtering

An estimate of the transform of the original image

$$\hat{F}(u,v) = \frac{G(u,v)}{H(u,v)}$$

$$\hat{F}(u,v) = \frac{F(u,v)H(u,v)+N(u,v)}{H(u,v)} = F(u,v) + \frac{N(u,v)}{H(u,v)}$$

1. We can't exactly recover the undegraded image because $N(u,v)$ is not known.
2. If the degradation function has zero or very small values, then the ratio $N(u,v)/H(u,v)$ could easily dominate the estimate $\hat{F}(u,v)$.

Minimum Mean Square Error (Wiener) Filtering

- **N. Wiener (1942)**
- **Objective**

  Find an estimate of the uncorrupted image such that the mean square error between them is minimized

$$e^2 = E\left\{(f - f)^2\right\}$$

The minimum of the error function is given in the frequency domain by the expression

$$F(u,v) = \left[\frac{H^*(u,v) S_f(u,v)}{S_f(u,v) |H(u,v)|^2 + S_\eta(u,v)} \right] G(u,v)$$

$$= \left[\frac{H^*(u,v)}{|H(u,v)|^2 + S_\eta(u,v) / S_f(u,v)} \right] G(u,v)$$

$$= \left[\frac{1}{H(u,v)} \frac{|H(u,v)|^2}{|H(u,v)|^2 + S_\eta(u,v) / S_f(u,v)} \right] G(u,v)$$

$$F(u,v) = \left[\frac{1}{H(u,v)} \frac{|H(u,v)|^2}{|H(u,v)|^2 + S_\eta(u,v)/S_f(u,v)} \right] G(u,v)$$

$H(u,v)$: degradation function

$H^*(u,v)$: complex conjugate of $H(u,v)$

$$|H(u,v)|^2 = H^*(u,v)H(u,v)$$

$S_\eta(u,v) = |N(u,v)|^2 =$ power spectrum of the noise

$S_f(u,v) = |F(u,v)|^2 =$ power spectrum of the undegraded image

$$F(u,v) = \left[\frac{1}{H(u,v)} \frac{|H(u,v)|^2}{|H(u,v)|^2 + K} \right] G(u,v)$$

K is a specified constant. Generally, the value of K is chosen interactively to yield the best visual results.

Measures of Quality

Singal-to-Noise Ratio (SNR)

$$SNR = \frac{\sum_{u=0}^{M-1}\sum_{v=0}^{N-1} |F(u,v)|^2}{\sum_{u=0}^{M-1}\sum_{v=0}^{N-1} |N(u,v)|^2}$$

This ratio gives a measure of the level of information bearing signal power to the level of noise power.

Mean Square Error (MSE)

$$\text{MSE}=\frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1}\left[f(x,y)-f(x,y)\right]^2$$

Root-Mean-Sqaure-Error (RMSE)

$$\text{RMSE}=\frac{\sum_{u=0}^{M-1}\sum_{v=0}^{N-1}f(x,y)^2}{\sum_{u=0}^{M-1}\sum_{v=0}^{N-1}|f(x,y)-f(x,y)|^2}$$

Constrained Least Squares Filtering

- In Wiener filter, the power spectra of the undegraded image and noise must be known. Although a constant estimate is sometimes useful, it is not always suitable.

- Constrained least squares filtering just requires the mean and variance of the noise.

$$F(u,v)=\left[\frac{H^*(u,v)}{|H(u,v)|^2+\gamma|P(u,v)|^2}\right]G(u,v)$$

$P(u,v)$ is the Fourier transform of the function

$$p(x,y)=\begin{bmatrix}0 & -1 & 0\\ -1 & 4 & -1\\ 0 & -1 & 0\end{bmatrix}$$

γ is a parameter

Geometric Mean Filter

$$F(u,v)=\left[\frac{H^*(u,v)}{|H(u,v)|^2}\right]^{\alpha}\left[\frac{|H(u,v)|^2}{|H(u,v)|^2+\beta\left[S_\eta(u,v)/S_f(u,v)\right]}\right]^{1-\alpha}G(u,v)$$

$\alpha=1$: inverse filter

$\alpha=0$: parametric Wiener filter

$\alpha=1/2$: geometric mean filter

Image Reconstruction

Image reconstruction in CT is a mathematical process that generates tomographic images from X-ray projection data acquired at many different angles around the patient. Image reconstruction has fundamental impacts on image quality and therefore on radiation dose.

Image Reconstruction from Projection

- Reconstruct an image from a series of projections

X-ray computed tomography (CT)

Computed tomography is a medical imaging method employing tomography where digital geometry processing is used to generate a three-dimensional image of the internals of an object from a large series of two-dimensional X-ray images taken around a single axis of rotation.

Backprojection: In computed tomography or other imaging techniques requiring reconstruction from multiple projections, an algorithm for calculating the contribution of each voxel of the structure to the measured ray data, to generate an image; the oldest and simplest method of image reconstruction.

Image Reconstruction: Example

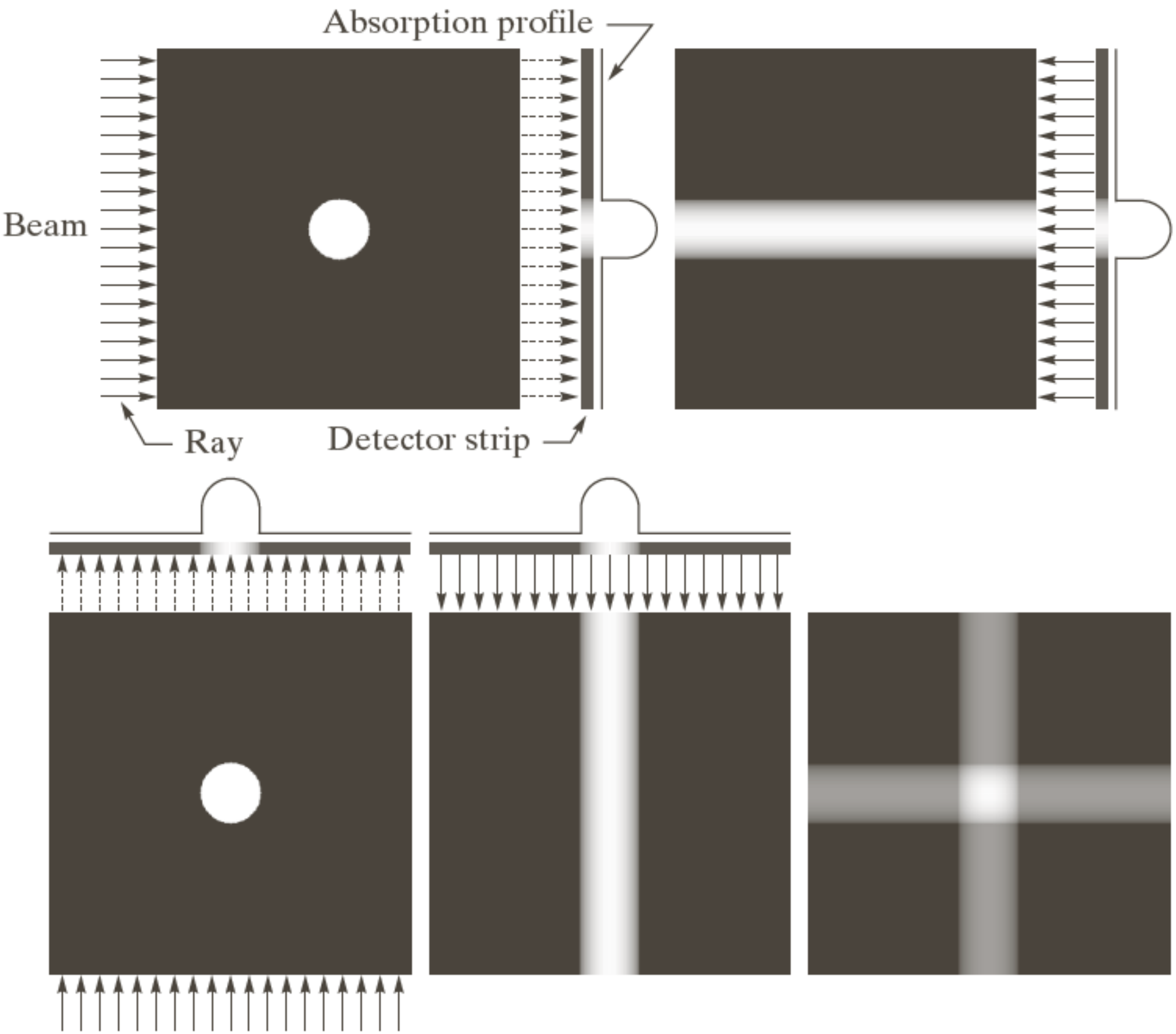
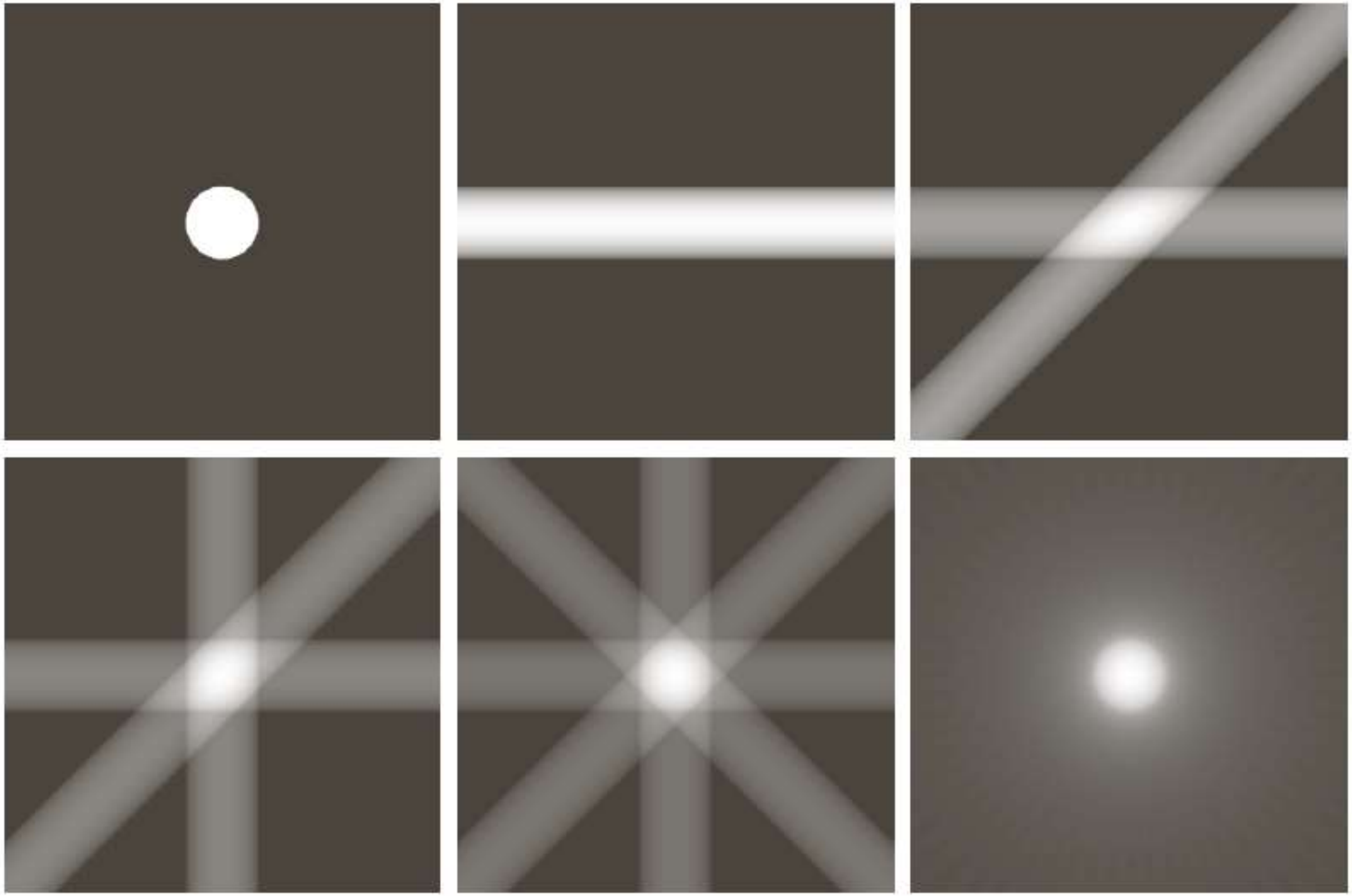


FIGURE 5.32 (a) Flat region showing a simple object, an input parallel beam, and a detector strip. (b) Result of back-projecting the sensed strip data (i.e., the 1-D absorption profile). (c) The beam and detectors rotated by 90°. (d) Back-projection. (e) The sum of (b) and (d). The intensity where the back-projections intersect is twice the intensity of the individual back-projections.



a	b	c
d	e	f

FIGURE 5.33
(a) Same as Fig. 5.32(a).
(b)–(e) Reconstruction using 1, 2, 3, and 4 backprojections 45° apart.
(f) Reconstruction with 32 backprojections 5.625° apart (note the blurring).

Other CTs

- **Electron beam CT** (Fifth-generation CT)

Electron beam tomography (EBCT) was introduced in the early 1980s, by medical physicist Andrew Castagnini, as a method of improving the temporal resolution of CT scanners.

High cost of EBCT equipment, and poor flexibility

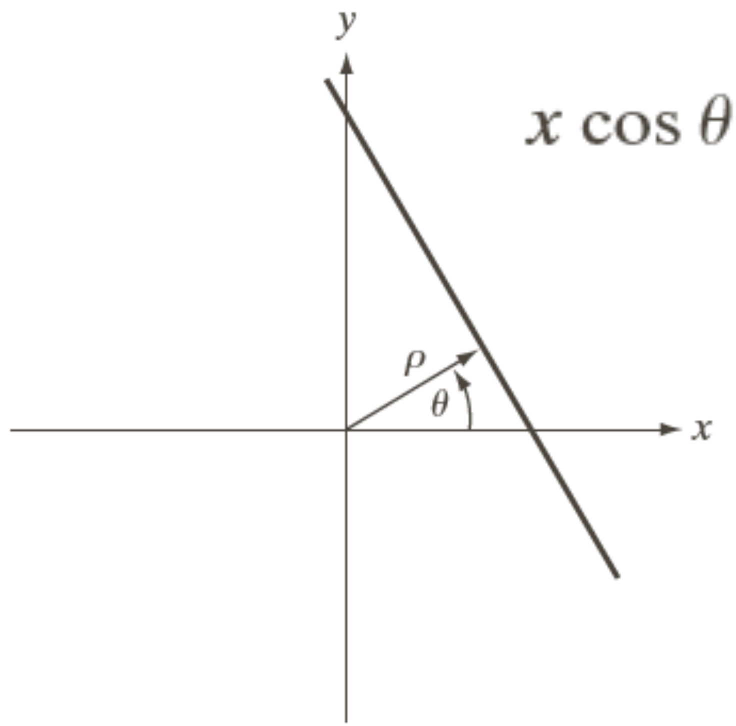
- **Helical (or spiral) cone beam computed tomography** (sixth-generation)

A type of three dimensional computed tomography (CT) in which the source (usually of x-rays) describes a helical trajectory relative to the object while a two dimensional array of detectors measures the transmitted radiation on part of a cone of rays emitting from the source

- **Multislice CT** (seventh-generation)

- The major benefit of multi-slice CT
  - Significant increase in detail
  - Utilizes X-ray tubes more economically
  - Reducing cost and potentially reducing dosage

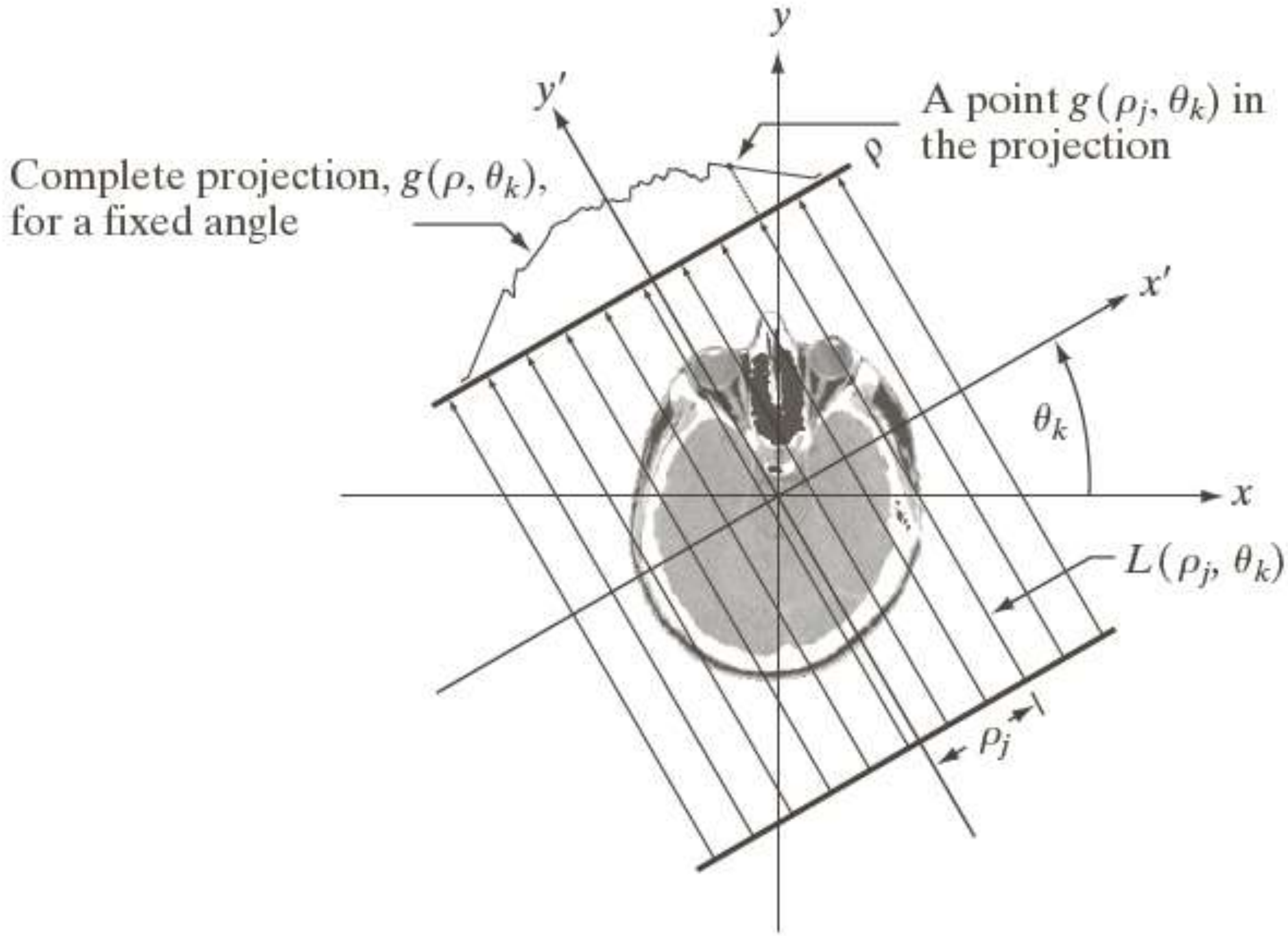
Projections and Radon Transform



$$x \cos\theta + y \sin\theta = \rho$$

FIGURE 5.36 Normal representation of a straight line.

FIGURE 5.37
Geometry of a parallel-ray beam.



$$g(\rho_j, \theta_k) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x, y)\delta(x\cos\theta_k + y\sin\theta_k - \rho_j)dxdy$$

- **Radon transform** gives the projection (line integral) of f(x,y) along an arbitrary line in the xy-plane

$$\Re\{f\} = g(\rho,\theta) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\delta(x\cos\theta + y\sin\theta - \rho)dxdy$$

$$\Re\{f\} = g(\rho,\theta) = \sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)\delta(x\cos\theta + y\sin\theta - \rho)$$

Example: Using the Radon transform to obtain the projection of a circular region

- Assume that the circle is centered on the origin of the xy-plane. Because the object is circularly symmetric, its projections are the same for all angles, so we just check the projection for $\theta = 0^\circ$

$$f(x,y) = \begin{cases} A & x^2 + y^2 \leq r^2 \\ 0 & \text{otherwise} \end{cases}$$

$$g(\rho,\theta)=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\delta(x\cos\theta+y\sin\theta-\rho)dxdy$$

$$=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\delta(x-\rho)dxdy$$

$$=\int_{-\infty}^{\infty} f(\rho,y)dy$$

$$=\int_{-\sqrt{r^2-\rho^2}}^{\sqrt{r^2-\rho^2}} f(\rho,y)dy$$

$$=\int_{-\sqrt{r^2-\rho^2}}^{\sqrt{r^2-\rho^2}} A dy$$

$$=\begin{cases} 2A\sqrt{r^2-\rho^2} & |\rho|\le \mathrm{r} \\ 0 & \text{otherwise} \end{cases}$$

$$g(\rho) = \begin{cases} 2A\sqrt{r^2 - \rho^2} & |\rho| \leq \mathrm{r} \\ 0 & \text{otherwise} \end{cases}$$

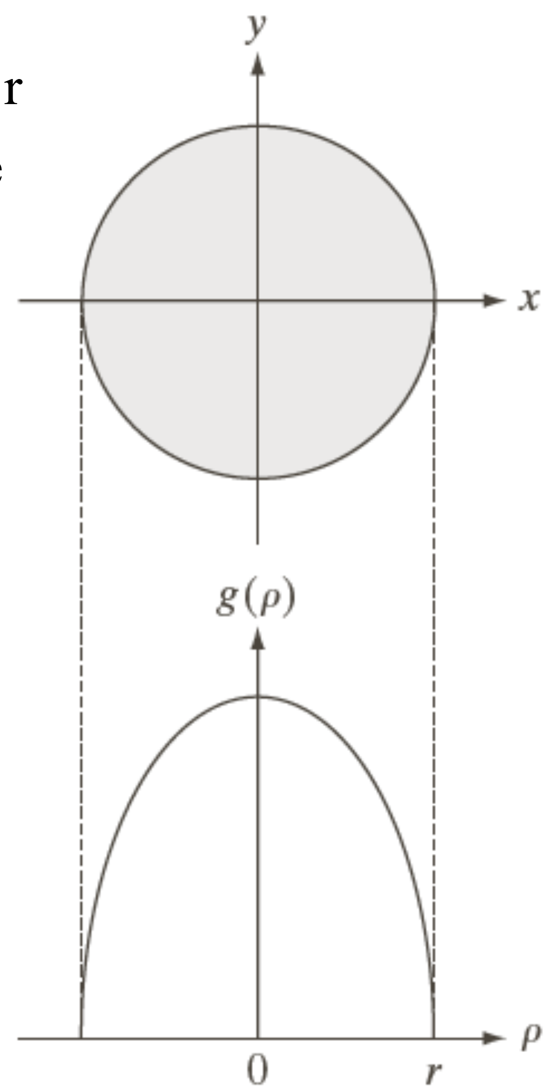


FIGURE 5.38 A disk and a plot of its Radon transform, derived analytically. Here we were able to plot the transform because it depends only on one variable. When g depends on both ρ and θ, the Radon transform becomes an image whose axes are ρ and θ, and the intensity of a pixel is proportional to the value of g at the location of that pixel.

Sinogram: Result of Radon Transform

- Sinogram: the result of Radon transform is displayed as an image with ρ and θ as rectilinear coordinates

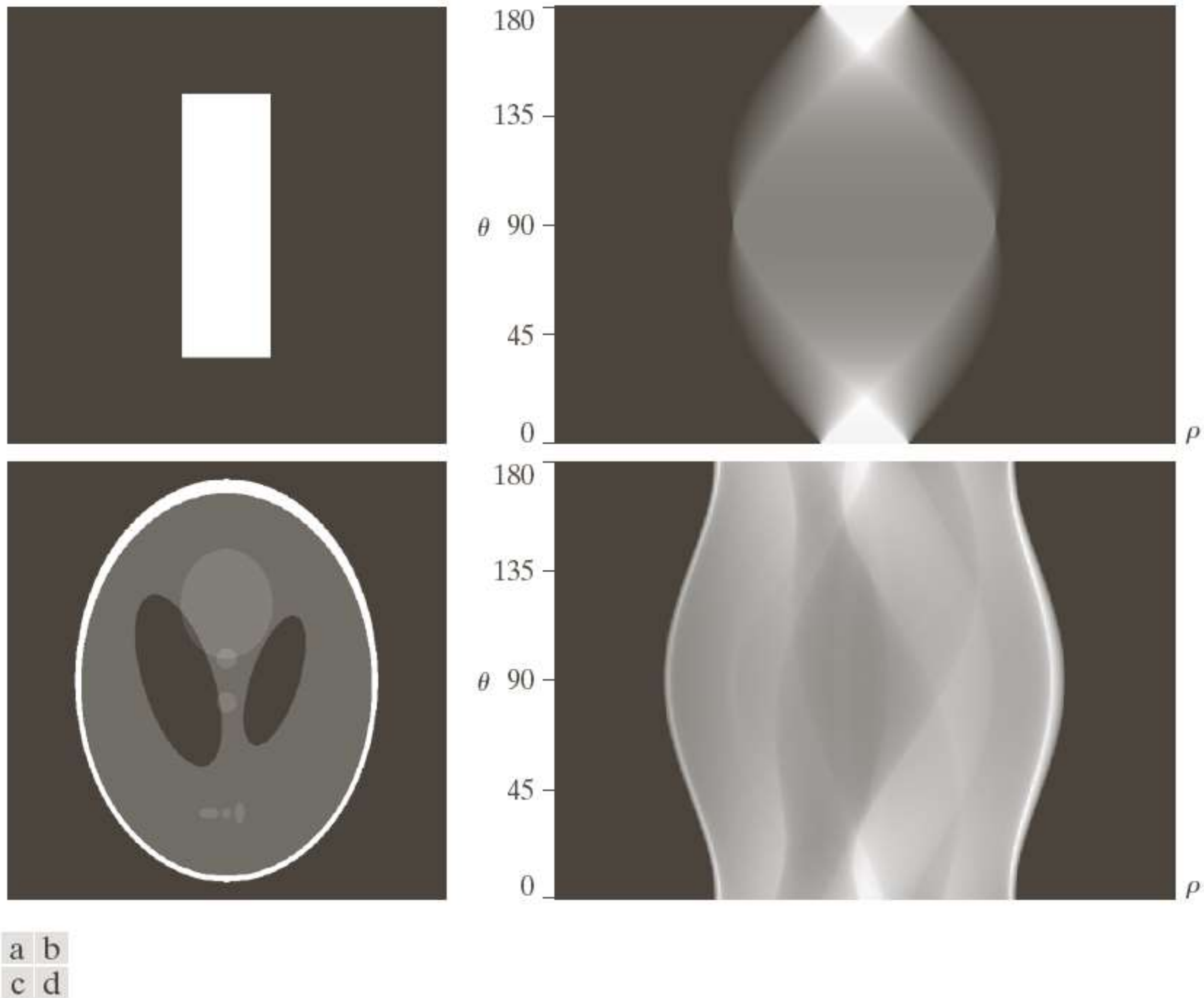


FIGURE 5.39 Two images and their sinograms (Radon transforms). Each row of a sinogram is a projection along the corresponding angle on the vertical axis. Image (c) is called the *Shepp-Logan phantom*. In its original form, the contrast of the phantom is quite low. It is shown enhanced here to facilitate viewing.

Image Reconstruction

$$f_\theta(x, y) = g(x\cos\theta + y\sin\theta, \theta)$$

$$f(x, y) = \int_0^\pi f_\theta(x, y)d\theta$$

$$f(x, y) = \sum_{\theta=0}^{\pi} f_\theta(x, y)$$

A back-projected image formed is referred to as a laminogram

Reconstruction Using Parallel-Beam Filtered Backprojections

$$f(x,y)=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}F(u,v)e^{j2\pi(ux+vy)}dudv$$

Let $u = w\cos\theta, v = w\sin\theta$, then $dudv = wdwd\theta$,

$$f(x,y)=\int_{0}^{2\pi}\int_{0}^{\infty}F(w\cos\theta, w\sin\theta)e^{j2\pi w(x\cos\theta+y\sin\theta)}wdwd\theta$$

$$=\int_{0}^{2\pi}\int_{0}^{\infty}G(w,\theta)e^{j2\pi w(x\cos\theta+y\sin\theta)}wdwd\theta$$

$$G(w,\theta+180^\circ)=G(-w,\theta)$$

$$f(x,y)=\int_{0}^{\pi}\int_{-\infty}^{\infty}|w|G(w,\theta)e^{j2\pi w(x\cos\theta+y\sin\theta)}dwd\theta$$

$$f(x,y)=\int_0^{\pi}\int_{-\infty}^{\infty}|w|\,G(w,\theta)e^{j2\pi w(x\cos\theta+y\sin\theta)}dw\,d\theta$$

$$=\int_0^{\pi}\left[\int_{-\infty}^{\infty}|w|\,G(w,\theta)e^{j2\pi w\rho}dw\right]_{\rho=x\cos\theta+y\sin\theta}d\theta$$

It's not integrable

Approach:

Window the ramp so it becomes zero outside of a defined frequency interval. That is, a window band-limits the ramp filter.

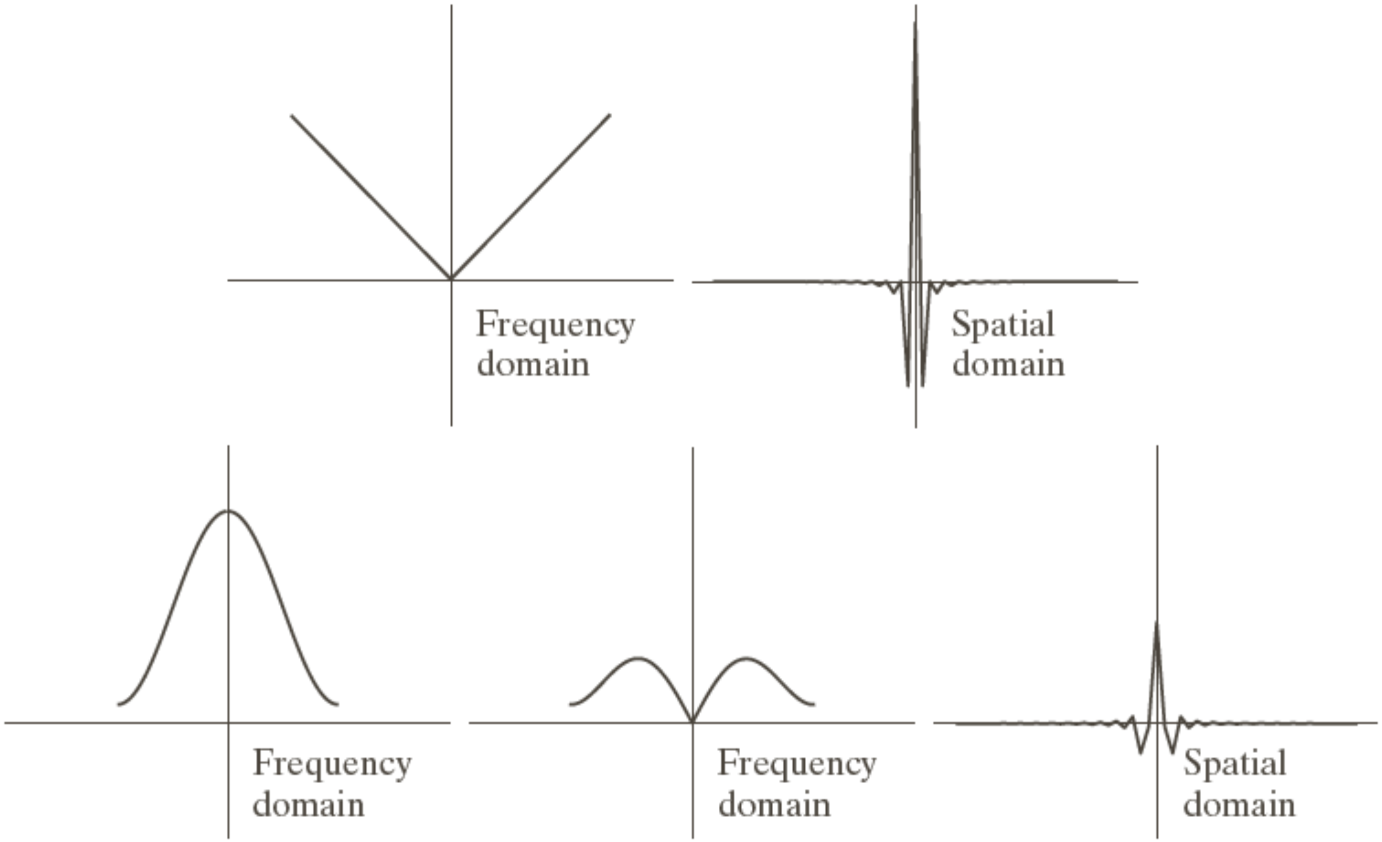
Hamming / Hann Window

$$h(w)=\begin{cases} c+(c-1)\cos\dfrac{2\pi w}{M-1} & 0\le w\le (M-1) \\ 0 & \text{otherwise} \end{cases}$$

$c=0.54,$ the function is called the <u>Hamming window</u>

$c=0.5,$ the function is called the <u>Han window</u>

Plot of Hamming Window



a b
c d e

FIGURE 5.42
(a) Frequency domain plot of the filter $|\omega|$ after band-limiting it with a box filter. (b) Spatial domain representation. (c) Hamming windowing function. (d) Windowed ramp filter, formed as the product of (a) and (c). (e) Spatial representation of the product (note the decrease in ringing).

Filtered Backprojection

The complete, filtered backprojection (to obtain the reconstructed image f(x,y)) is described as follows:

1. Compute the 1-D Fourier transform of each projection
2. Multiply each Fourier transform by the filter function |w| which has been multiplied by a suitable (e.g., Hamming) window
3. Obtain the inverse 1-D Fourier transform of each resulting filtered transform
4. Integrate (sum) all the 1-D inverse transforms from step 3

Questions???

Thank You!!!